

ECO387C - Fall 2024

Discussion 6

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1 Introduction

Recall the stochastic neoclassical growth model, where the planner solves:

$$\max_{\{c_t, k_{t+1}\}_{t=0}^{\infty}} \mathbb{E}_0 \left\{ \sum_{t=0}^{\infty} \beta^t u(c_t) \right\}$$

subject to

$$\begin{aligned} c_t + k_{t+1} &\leq z_t f(k_t) + (1 - \delta)k_t \\ \log(z_{t+1}) &= \rho \log(z_t) + \varepsilon_{t+1}, \quad \varepsilon \sim \text{I.I.D. } \mathcal{N}(0, \sigma_\varepsilon) \end{aligned}$$

for a given k_0 and z_0 . The corresponding Bellman equation is given by:

$$v(k, z) = \max_{\{0 \leq k' \leq z f(k) + (1 - \delta)k\}} \{u(z f(k) + (1 - \delta)k - k') + \beta \mathbb{E}_{z'|z} v(k', z')\}$$

for all (k, z) , where z' evolves according to the AR(1) process as above.

We can still use value function iteration to solve this, but solving this problem now requires us to compute expectations, or $\mathbb{E}_{z'|z} v(k', z')$. One way is to approximate the AR(1) process, which is continuous, with a **finite-state Markov chain**.

2 Discretizing AR(1) processes

2.1 Tauchen (1986)

Tauchen's method allows us to find a probability matrix that governs the transition on a given discrete grid that approximates an AR(1) process ¹

¹Tauchen, George. "Finite state markov-chain approximations to univariate and vector autoregressions." *Economics letters* 20.2 (1986): 177-181.

Suppose that $\varepsilon'_{i,t} \sim N(0, 1)$, then I can write an AR(1) process as

$$z' = (1 - \rho)\bar{z} + \rho z + \sigma \varepsilon'$$

$$\varepsilon' = \frac{z' - \rho z - (1 - \rho)\bar{z}}{\sigma}$$

Tauchen's method is based on the fact that conditional on z , z' is distributed normally with mean $(1 - \rho)\bar{z} + \rho z$ and the standard deviation of σ . Now let $p_{jk} = P(z' = k | z = j)$ be the transition from value j to value k . Let further w denote difference between two neighboring gridpoints on a equidistant grid. The grid is denoted by $z_1, \dots, z_{N_z}$. Then, for $k \in \{2, \dots, N_z - 1\}$. Note that Φ denotes the cdf of standard normal distribution.

$$\begin{aligned} p_{jk} &= P\left(z_k - \frac{w}{2} \leq (1 - \rho)\bar{z} + \rho z_j + \sigma \varepsilon' \leq z_k + \frac{w}{2}\right) \\ &= P\left(\frac{z_k - \frac{w}{2} - (1 - \rho)\bar{z} - \rho z_j}{\sigma} \leq \varepsilon' \leq \frac{z_k + \frac{w}{2} - (1 - \rho)\bar{z} - \rho z_j}{\sigma}\right) \\ &= \Phi\left(\frac{z_k + \frac{w}{2} - (1 - \rho)\bar{z} - \rho z_j}{\sigma}\right) - \Phi\left(\frac{z_k - \frac{w}{2} - (1 - \rho)\bar{z} - \rho z_j}{\sigma}\right) \end{aligned}$$

For $k = 1$

$$\begin{aligned} p_{j1} &= P\left((1 - \rho)\bar{z} + \rho z_j + \sigma \varepsilon' \leq z_1 + \frac{w}{2}\right) \\ &= P\left(\varepsilon' \leq \frac{z_1 + \frac{w}{2} - (1 - \rho)\bar{z} - \rho z_j}{\sigma}\right) \\ &= \Phi\left(\frac{z_1 + \frac{w}{2} - (1 - \rho)\bar{z} - \rho z_j}{\sigma}\right) \end{aligned}$$

For $k = N_z$

$$\begin{aligned} p_{jN_z} &= P\left((1 - \rho)\bar{z} + \rho z_j + \sigma \varepsilon' \geq z_{N_z} - \frac{w}{2}\right) \\ &= 1 - P\left(\frac{z_{N_z} - \frac{w}{2} - (1 - \rho)\bar{z} - \rho z_j}{\sigma} \leq \varepsilon'\right) \\ &= 1 - \Phi\left(\frac{z_{N_z} - \frac{w}{2} - (1 - \rho)\bar{z} - \rho z_j}{\sigma}\right) \end{aligned}$$

Summary For all $j \in \{1, \dots, N_z\}$, the transition probabilities p_{jk} are given by:

$$p_{jk} = \begin{cases} \Phi\left(\frac{z_k + \frac{w}{2} - (1 - \rho)\bar{z} - \rho z_j}{\sigma}\right) & \text{if } k = 1, \\ \Phi\left(\frac{z_k + \frac{w}{2} - (1 - \rho)\bar{z} - \rho z_j}{\sigma}\right) - \Phi\left(\frac{z_k - \frac{w}{2} - (1 - \rho)\bar{z} - \rho z_j}{\sigma}\right) & \text{if } k \in \{2, \dots, N_z - 1\}, \\ 1 - \Phi\left(\frac{z_{N_z} - \frac{w}{2} - (1 - \rho)\bar{z} - \rho z_j}{\sigma}\right) & \text{if } k = N_z. \end{cases}$$

2.2 Rouwenhorst (1995)

Consider the same AR(1) process:

$$z' = (1 - \rho)\bar{z} + \rho z + \sigma \varepsilon'$$

where $|\rho| < 1$, and ε is an i.i.d. shock that follows a standard normal distribution. This AR(1) process has a stationary normal distribution with mean $\bar{z}$ and variance $\sigma_z^2 = \frac{\sigma^2}{1-\rho^2}$. Rouwenhorst (1995) proposes a discretization method that matches the three key moments of the AR(1) process: the stationary mean, variance, and autocorrelation.² The method follows the following three steps.

First, define $v = \sqrt{\frac{n-1}{1-\rho^2}}\sigma$ and create n discrete grid points over the interval $[\bar{z} - v, \bar{z} + v]$. n is going to determine the number of states of the Markov process.

Second, construct the Rouwenhorst probability transition matrix recursively. For $n = 2$, the transition matrix is:

$$P_2 = \begin{bmatrix} p & 1-p \\ 1-q & q \end{bmatrix}$$

For $n > 2$, the P_n matrix can be generated recursively from P_{n-1} as follows:

$$p \begin{bmatrix} P_{n-1} & \mathbf{0} \\ \mathbf{0}' & 0 \end{bmatrix} + (1-p) \begin{bmatrix} \mathbf{0} & P_{n-1} \\ \mathbf{0}' & 0 \end{bmatrix} + p \begin{bmatrix} \mathbf{0}' & \mathbf{0} \\ P_{n-1} & 0 \end{bmatrix} + (1-p) \begin{bmatrix} 0 & \mathbf{0} \\ \mathbf{0}' & P_{n-1} \end{bmatrix}$$

where $\mathbf{0}$ is an $(n-1) \times 1$ column vector of zeros. Divide all but the top and bottom rows by 2 so that the sum of the elements of each row equals 1. For example, P_3 is:

$$P_3 = \begin{bmatrix} p^2 & 2p(1-p) & (1-p)^2 \\ p(1-q) & pq + (1-p)(1-q) & q(1-p) \\ (1-q)^2 & 2q(1-q) & q^2 \end{bmatrix}$$

This construction method is particularly useful for highly persistent processes ($\rho > 0.9$).³ It also matches the key moments of the AR(1) process *regardless of the number of grid-points*. For example, it can be shown that $p + q - 1 = \rho$. In most applications, it is assumed that $p = q$, and thus, we can simply set $p = \frac{\rho+1}{2}$ to construct the P_n . Furthermore, it can also be shown that $\frac{v^2}{n-1} = \frac{\sigma^2}{1-\rho^2}$. So setting $v = \sqrt{\frac{n-1}{1-\rho^2}}\sigma$ as done in the first step ensures that the constructed Markov transition matrix matches the variance of the underlying AR(1) process.

²Rouwenhorst, K. Geert. "Asset pricing implications of equilibrium business cycle models." *Frontiers of Business Cycle Research*, 1 (1995): 294-330.

³Kopecky, Karen A., and Richard MH Suen. "Finite state Markov-chain approximations to highly persistent processes." *Review of Economic Dynamics*, 13.3 (2010): 701-714.

3 National Income and Product Accounts (NIPA)

3.1 Background

Accounts provide comprehensive measurement of production, income, and expenditure for the U.S. economy. Developed in the the 1930s and 1940s by the Department of Commerce. Currently, the accounts are constructed quarterly by government economists and statisticians in the Bureau of Economic Analysis (BEA) (website: www.bea.gov). In constructing the accounts, the BEA works in collaboration with the Census Bureau (<http://www.census.gov/>) and the Bureau of Labor and Statistics (<http://www.bls.gov/>).

3.2 Example

Suppose that an economy has three stages of production:

- Stage 1: mines & labor & machinery $\Rightarrow$ iron ore
- Stage 2: ore & labor & machinery $\Rightarrow$ steel
- Stage 3: steel & labor & machinery $\Rightarrow$ autos & new machines

Definition: The **Gross Domestic Product (GDP)** of an economy is the value of all final goods and services made within a country during a specific period (quarter / year).

Terms

- "Factors of Production (FOP's):" inputs from nature and long-lived machinery (in our example, labor, mines, machinery).
- "Intermediate Goods:" products that are quickly used up in later stages (in our example iron ore, steel).
- "Final Goods:" final output - not used in subsequent production (in our example autos and new machines, but the following period, new machines become FOP's).

3.3 Construction of GDP

There are three ways to construct GDP. All three are equivalent and allow us to view GDP from different perspectives.

1. Production Approach: GDP is the current market value of all *newly produced* goods and services in the economy during a fixed period of time. It can be measured by adding the *value added* over all stages of production:

$$\begin{aligned}
 GDP_t &= \sum_{\text{all stages}} [\text{value added}] \\
 &= \sum_{\text{all stages}} [\text{value of stage's production} - \text{value of intermediates used}] \\
 &= \underbrace{[\text{ore produced}]}_{\text{Stage 1}} + \underbrace{[\text{steel produced} - \text{ore used}]}_{\text{Stage 2}} + \underbrace{[\text{autos \& machines produced} - \text{steel used}]}_{\text{Stage 3}}
 \end{aligned}$$

Note that we include both sold and unsold (increment to inventory) output for the value of the stage's production. Also, note that if we just took ore + steel + autos & new machines we would be double counting (for example, stage 1 ore produced and the ore used in steel production).

2. Expenditure Approach: GDP is the sum of spending on all final goods and services production in the economy.

$$\begin{aligned}
 GDP_t &= C_t + I_t + G_t + NX_t \\
 &= \text{the value of final goods produced and sold} \\
 &\quad + \text{the value of increments (changes) to business inventories} \\
 &= [\text{autos and new machines produced and sold}] \\
 &\quad + [\text{ore, steel, autos and new machines produced but unsold}]
 \end{aligned}$$

1. **Consumption** (C_t) is spending by domestic households on final goods and services including those produced abroad. It is the largest expenditure component, usually accounting for about two-thirds of GDP in the United States. Consumption expenditures are grouped into three categories:

- *Consumer durables*: Long-lived consumer items, such as automobiles, electronic goods, and major appliances (but not houses, which are classified under investment).
- *Nondurable goods*: Shorter-lived items, such as food, clothing, and fuel.
- *Services*: Education, health care, financial services, and transportation.

2. **Investment** (I_t) includes both spending for new capital goods, called *fixed investment*, and changes in firms' inventory holdings, called *inventory investment*. Fixed investment in turn has two major components: non-residential and residential investment.

- *Non-residential investment*: Spending by businesses on structures (factories, warehouses, and office buildings), equipment (such as machines, vehicles, and furniture), and intellectual property products (R&D).
- *Residential investment*: Spending on new houses and apartment. Houses and apartment buildings are treated as capital goods because they provide a service (shelter) over a long period of time. Note that the purchase of existing housing does not count because they were produced in earlier periods.
- *Inventory investment*: Change in inventories held by firms. If inventories are increasing, inventory investment is positive, but if they are decreasing, inventory investment is negative.

3. **Government purchases** (G_t) include any expenditure by the government for a currently produced good or service, foreign or domestic. While the National Income and Product Accounts (NIPA) classify these purchases at the federal and state (local) levels, government spending can also be categorized into the following types:⁴

- Government consumption: Purchases for short-lived goods and services like health care and police.
- Government investment: Spending for capital goods like buildings and computers.

However, not all checks written by the government are for purchases of goods and services. **Transfers**, which include government payments for Social Security and Medicare benefits, unemployment insurance, welfare payments, and so on, and **Interest payments on government debt** are payments by the government that are not made in exchange for current goods and services. Hence, they are excluded from the government purchases category and are not counted in GDP.

4. **Net Exports** (NX_t) are exports minus imports. **Exports** are the goods and services produced within a country that are purchased by foreigners. **Imports** are the goods and services produced abroad that are purchased by a country's residents.

Examples:

1. French tourist buys a hamburger in Ann Arbor and eats it: US Y increases through the rise in NX because the burger was produced in the US and exports have risen.

⁴One of the key topics regarding government purchases is fiscal multiplier. Chris has a paper on fiscal multipliers depending on the type of government expenditure: Boehm, Christoph E. "Government consumption and investment: Does the composition of purchases affect the multiplier?." *Journal of Monetary Economics* 115 (2020): 80-93.

French Y stays still because the increase in C is offset by a fall in NX . Note that accounts balance for each country (identity (A) holds).

$$\begin{aligned} Y_t^{\text{US}} \uparrow &= C_t^{\text{US}} + I_t^{\text{US}} + G_t^{\text{US}} + NX_t^{\text{US}} \uparrow \\ Y_t^{\text{France}} &= C_t^{\text{France}} \uparrow + I_t^{\text{France}} + G_t^{\text{France}} + NX_t^{\text{France}} \downarrow \end{aligned}$$

2. French business buys a new machine made in the US and produces with it in the US: US Y rises through the rise in I because the new machine counts in production and the new machine counts in investment. France's national accounts do not change since the investment is not made on French soil.

$$\begin{aligned} Y_t^{\text{US}} \uparrow &= C_t^{\text{US}} + I_t^{\text{US}} \uparrow + G_t^{\text{US}} + NX_t^{\text{US}} \\ Y_t^{\text{France}} &= C_t^{\text{France}} + I_t^{\text{France}} + G_t^{\text{France}} + NX_t^{\text{France}} \end{aligned}$$

3. Auto produced in the US in November 2008. Sold to a US resident in February 2009: US (2008): Y rises, I rises. US(2009): C rises, I falls (because the car is taken out of inventories).

3. Income Approach: GDP is the sum of payments to factors of production (after tax) along with indirect taxes:

$$GDP_t = \sum_{\text{all stages}} [\text{payments to factors of production} + \text{indirect taxes}]$$

A key part of the income approach is a concept known as **National income**, which is the sum of the following types of income:

- *Compensation of employees:* Income of workers (excluding the self-employed). It includes wages, salaries, employee benefits, and employer contributions to Social Security.
- *Proprietors' income:* Income of the self-employed. Because many self-employed people own some capital, proprietors' income includes both labor income and capital income.
- *Rental income of persons:* Income earned by individuals who own land or structures that they rent to others.
- *Corporate profits:* Profits earned by corporations and represent the remainder of corporate revenue after wages, interest, rents, and other costs have been paid.
- *Net interest and miscellaneous payments:* **Net interest** reflects the net income earned by the government and persons for providing certain financial resources to the production process. The measure includes only the interest that is charged for

the provision of financial resources used in, or associated with production. **Miscellaneous payments** are a measure of the rent and royalty payments that accrue to federal to state and local government owners of land or subsoil assets in return for leasing the rights to these assets for various productive uses. This includes federal government receipts from leasing federally-owned lands for mining or energy production and state and local receipts from allowing temporary possession of government-owned buildings, land, or other fixed properties (grazing fees, timberland leases, and natural resource exploration and production).

- *Taxes on production and imports:* This consists of taxes payable on products when they are produced, delivered, sold, transferred, or otherwise disposed of by their producers (including federal excise taxes, customs duties, and state and local sales taxes) and of other taxes on production, such as taxes on ownership of assets used in production (including local real estate taxes). These taxes do not include taxes on income or profits.

3.4 Identities

(A) Disposition of GDP (via expenditure approach):

$$Y_t = C_t + I_t + G_t + NX_t$$

(B) Household budget constraint

$$Y_t + TR_t + NFP_t + NTR_t = C_t + T_t + S_t$$

- TR_t : transfer from government.
- NFP_t : net factor payments from abroad to US residents, e.g. shares of stock in UK - dividends.
- NTR_t : net transfer payments from abroad (remittances), e.g. US citizen works in France and gets paid by the French; sends some money home back to his family.
- T_t : taxes paid in US (indirect taxes and income taxes).
- S_t : domestic private savings.

Note, in this class we predominantly deal with closed economies without government. This identity then becomes:

$$Y_t = C_t + S_t$$

(C) Government budget identity:

$$\underbrace{T_t + \dot{M}_t + \dot{B}_t^G}_{\text{Inflow}} = \underbrace{G_t + TR_t}_{\text{Outflow}}$$

- M_t is the stock of US currency outside of government (in \$'s). $\dot{M}_t$ is seignorage income. The government can choose to print more money and is thus in essence borrowing interest-free. Further, printing additional currency can increase inflation and thus reduce the government's liabilities. Note that $\dot{M}_t$ is a flow variable:

$$\dot{M}_t = \frac{dM_t}{dt} = \lim_{\Delta \rightarrow 0} \frac{M_{t+\Delta} - M_t}{\Delta} = \$/\text{year}$$

- B_t^G is the stock of government bonds outside of the government. $\dot{B}_t^G$ is simply the time derivative of B_t^G i.e. $\dot{B}_t^G = \frac{dB_t^G}{dt}$.

We will often use identity (C) as a constraint in an optimization problem when we introduce the government into our models.

(D) Balance of Payments Identity

$$\underbrace{X_t + NFP_t + NTR_t + CF_t}_{\text{Outflow of \$ from FEM}} = \underbrace{IM_t}_{\text{Inflow of \$ from FEM}}$$

- CF_t : net foreign financial investment in the US. This includes any type of US asset that a foreigner owns, for example, bonds, money, or stocks. In essence, there are two ways to fund imports (ignoring NFP_t and NTR_t).
- One way is via goods (X_t) and one way is via issuing claims on future output of the US.
- NX_t is broken into its components, X_t (exports) and IM_t (imports).

This identity has to do with the Foreign Exchange Market (FEM). Since the FEM has no way of making new dollars, the dollars that are taken out of the FEM must equal the dollars that are put into the FEM. On the left-hand side of (D), we have all the dollars that are taken out of the FEM. For example, if a US resident exports to Japan, the foreigner must put yen on the FEM and withdraw dollars in order to pay the US exporter. On the other hand, if US residents import something from Japan, they must put dollars into the FEM and withdraw yen to pay the Japanese exporter.

(E) Saving/Investment Accounting Identity:

$$C_t + T_t + S_t = Y_t + TR_t + NFP_t + NTR_t \quad \text{by (B)}$$

$$\Leftrightarrow C_t + T_t + S_t = C_t + I_t + G_t + X_t - IM_t + TR_t + NFP_t + NTR_t \quad \text{by (A)}$$

$$T_t + S_t = I_t + G_t - CF_t + TR_t \quad \text{by (D)}$$

$$\Leftrightarrow I_t = S_t + \underbrace{(T_t - G_t - TR_t)}_{\text{Government budget surplus}} + CF_t$$

This identity states that investment has to be financed somehow: either via private domestic savings, government expenses, or foreign investment. With a closed economy and no government, this is simply:

$$S_t = I_t$$

3.5 Consumer Price Index (CPI)

Definition If P_t is the price level in period t and P_{t+1} is the price level in period $t + 1$, the **rate of inflation** between t and $t + 1$, or π_{t+1} , is

$$\pi_{t+1} = \frac{P_{t+1} - P_t}{P_t} = \frac{\Delta P_{t+1}}{P_t}$$

The CPI helps us measure the rate of inflation, as well as the cost of living. It is calculated by the Bureau of Labor and Statistics on a monthly basis. CPI is constructed by calculating the price of a fixed list, or "basket," of consumer goods and services, including many specific items of food, clothing, housing and fuel. The CPI is then calculated as the current cost of the basket of consumer items divided by the cost of the same basket of items in the base year.

There are two classic problems with the CPI.

- *Quality adjustment bias*: increases in price due to increases in quality can be incorrectly interpreted as inflation because changes in quality are hard to quantify.
- *Substitution bias*: the CPI assumes that consumers purchase a basket of goods and services that is fixed over time, ignoring the fact that consumers can (and do) substitute cheaper goods or services for more expensive ones.

As a result of both these effects, the CPI tends to overstate the actual rate at which cost of living rises.